Carlos Gafa’

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EDUCATION

University of Malta

Micro-credential in Intelligent Algorithmic Trading (ARI5123) *Feb 2025 – Grade Pending*

- Project: Determining Entry/Exit Points of a Pairs Trading Strategy Using Online Reinforcement Learning
- Relevant Coursework: Reinforcement Learning (DQN, Policy Gradient, Actor-Critic Methods), Hidden Markov Models, Technical Analysis

Micro-credential in Financial Engineering (ARI5122) *Nov 2024 – Feb 2025*

- Relevant Coursework: PCA, Volatility Models (GARCH, GBM, RNNs), Factor Analysis and Investing

Master of Science in Engineering (Electrical) (By Research) *Oct 2023 – Jun 2025*

- Passed with Distinction
- In collaboration with the Paul Scherrer Institute and the University of Cambridge
- Relevant Skills: Optimization Algorithms, Inverse Analysis Methods, Data Analysis and Statistics

Bachelor of Science (Hons.) in Mathematics and Physics *Sep 2019 – Jul 2023*

- First Class Honours
- Dean’s List of Science
- Relevant Coursework: Python Simulations, Markov Chain Monte Carlo Methods, Probability

G.F. Abela – Junior College

Matriculation Certificate *Sep 2017 – Jul 2019*

- A’ Levels in: Mathematics (A), Biology (A), Chemistry (A)

WORK EXPERIENCE

PricewaterhouseCoopers — Malta

Data Analyst *Nov 2024 – Ongoing*

Working collaboratively within PwC's Data Analytics team on client projects, I gained experience in:

- Data Modelling: Developed, and optimized data models and robust ETL pipelines for data warehousing initiatives to enhance data quality and reporting performance.
- BI Reporting: Created insightful reports using tools such as Power BI to support strategic, data-driven decisions.
- Automation: Development of technology driven tools to automate interest rate risk calculations.
- Technologies Used: Python, SQL, Azure Data Factory, Power BI, Alteryx

Paul Scherrer Institute — Switzerland

Physicist *Sep 2023 – Nov 2024*

During my time with the Insertion Devices group at PSI, I played a key role in the High-Temperature Superconducting Undulator (HTSU) project, wherein:

- I improved FEM simulation efficiency by extending the H-ϕ formulation to cover ferromagnets,
- I developed a novel shimming algorithm to make iterative geometrical design changes targeted to decrease the phase error of the HTSU, integrating both simulation models and the latest experimental data at each step,
- I developed gradient based algorithms to optimize the magnetic field integrals of the HTSU,
- I applied advanced statistical analysis and inverse analysis methods to identify and characterize key HTS parameters, and predict time-series decay (flux creep) in superconductors. Predictive models were validated against experimental data.

Additionally, I spent three weeks at the University of Cambridge, where I assisted in upgrading the HTSU’s measurement system. My work was accepted as an oral contribution at the HTS Modelling Workshop 2024 and Swiss Physical Society Meeting 2024. Further work is pending publication.

PricewaterhouseCoopers **Malta**
Data Analyst *Jun 2022 – Sep 2023 (Seasonal)*

Peer Reviewed Publications

White Dwarf envelops and temperature corrections in exponential f(T) gravity

General Relativity and Gravitation *Dec 24, 2024*
DOI: *10.1007/s10714-024-03341-6*
Compact stars have long served as a test bed of gravitational models and their coupling with stellar matter. In this work, the Tolman-Oppenheimer-Volkoff equation was derived and solved for an exponential model in f(T) gravity using Salpeter's equation of state. To obtain a more realistic behaviour, inputs from standard white dwarf evolutionary models are used to calibrate the model for core temperature and envelope thickness. Finally, constraints on the models' parameters are obtained using Markov Chain Monte Carlo methods, which are comparable to results obtained using cosmological survey data. This consistency across the strong astrophysical and weak cosmological scales shows reasonable viability of the underlying model.

Experimental results of a YBCO bulk superconducting undulator magnetic optimization

Physical Review Accelerators and Beams *Oct 10, 2024*
DOI: *10.1103/PhysRevAccelBeams.27.100702*
The magnetic field optimization of RE-Ba-Cu-O (REBCO, RE=rare earth) bulk superconducting undulators is a fundamental step toward their implementation in an accelerator driven photon source, like a synchrotron or a free electron laser. In this article, we propose a statistical sorting algorithm to reduce the undulator's phase error based on the reconstruction of the trapped current inside the bulks of a staggered array undulator. The results obtained with a YBCO short prototype field-cooled down to 10 K in a 10 T magnetic field are reported. Finally, its performance is critically discussed in light of 2D magnetic field maps of its individual components, obtained at LN2 after the magnetization tests.

Professional Certifications

Microsoft

Microsoft Certified: Azure Data Fundamentals *Nov 2024*
Credential ID: *D9FE60CD73CB9710*